

Ship/HULL NO.	Ship/HULL NAME	JON:	ICN:	Rev:	Key Op:
LCU-1651	NBU-7		38NBA59303		A01

TITLE:

CHT Piping; chemically clean, inspect and replace

JOB SPECIFICATION

Anticipated Award Date: 12/08/2025 5/01/2026		Production Completion Date: 8/21/2026 8/29/2026		PREPARING ACTIVITY:	
POP Start Date: 2/02/2026 6/01/2026		POP End Date: 1/15/2027 9/30/2026		SASEBO	
PREPARED BY: A.HAMADA 4/9/2026 Avelino L. Reyes Jr.			CODE P HA	DATE 8/21/2025	
APPROVED BY: S.MURAKAMI <i>SM</i>			CODE S225		
CONCURRED BY: T.TAKAMASU <i>TT</i>			CODE S234		
ASSIST PLANNER'S NAME: K.KOMATSU			CODE H KO		
ASSIST PLANNER'S NAME:			CODE		
ASSIST PLANNER'S NAME:			CODE		

WORK TO BE DONE IN DRY-BERTHING

DESCRIPTION:

JSN(S): CU51-2062

CASREP #

S/F CONTACT: ORTEGA / KHULLAR GSCS

CWP #

CCNS #

Approved by S300/S900

A. REYES
PM, S300

Date:

9/29/25

Avl:

6C2

4/9/2026

Avelino L. Reyes Jr.

LWC:

Work Completed

S/Force: Rate: Date:

SHIPSUPT: Date: SRF CONTROL NO.: 1301357121

WORK CERTIFICATION SIGNATURES

COMPLETION OF WORK REVIEW:	BADGE NO.:	SHOP/CODE:	PHONE	DATE:
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1. SCOPE:

- 1.1 Title: CHT Piping; chemically clean, inspect and replace
- 1.2 Location of Work:
 - 1.2.1 FWD Engine Room (2-50-0-E)
 - 1.2.2 WR, WC, and SHR (1-44-1-L)
 - 1.2.3 Equipment Repair Parts and Ordnance Store Room (2-40-0-A)
- 1.3 Identification:
 - 1.3.1 Quantity (One EA), CHT Tank, (Capacity: 300 Gallon, CRES), Located in 1.2.1
 - 1.3.2 Quantity (Total 80 FT), 3/4 Inch IPS through 2 Inch IPS CHT Piping Located in 1.2, CuNi, See ENCLOSURE (A)
 - 1.3.3 Quantity (Total 10 FT), 3/4 Inch IPS through 2 Inch IPS CHT Piping (Part of CHT Piping Listed in 1.3.2) Which Leaks During Chemical Cleaning Process or Rejected by Ultrasonic Inspection, CuNi

2. REFERENCES:

- 2.1 Standard Items (FY 26)
- 2.2 500-5104327 Rev C, Vacuum Sewage Collection, Holding & Transfer System (VSCHTS)
- 2.3 S9086-T8-STM-010/CH-593/Rev 8, Pollution Control
- 2.4 MIL-STD-3026A, Chemical Cleaning of Sewage Collection Piping Systems on Navy Surface Ships
- 2.5 S9AA0-AB-GOS-010/GSO Rev 11, General Specifications for Overhaul of Surface Ships (GSO)
- 2.6 S9086-RK-STM-010/CH-505 Rev 6, Piping System

3. REQUIREMENTS:

(V) "REPORT"

- 3.1 Submit one legible copy of Marine Chemist certificate to SUPERVISOR including certified location.

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3.2 Accomplish the requirements of ENCLOSURE (T) for cleaning of each CHT tank listed in 1.3.1, using 2.2 for guidance, and clean piping affected by this work item to a gas/oil free condition, in accordance with 29 CFR 1915 Subpart B.

3.2.1 Disinfect each CHT tank and associated equipment in accordance with 2.3.

3.3 Chemically clean CHT piping system listed in 1.3.2 up to but not including the tank listed in 1.3.1 with RYDLYME or equal, using 2.4 and ENCLOSURE (A) for guidance.

3.3.1 Accomplish the requirement of 009-09 of 2.1 for process control procedure of how the chemical cleaning process of the CHT piping.

3.3.2 Cleanliness must be Level III of Section 505j2 of 2.5.

3.3.3 Disconnect and reconnect each mechanical joint on CHT piping to facilitate accomplishment of chemical cleaning.

3.3.4 Accomplish a flush of the CHT piping system, using clean fresh water for a minimum of one hour and ensure piping system is leak free. Allowable external leakage: None

3.3.5 Neutralize the flushing chemical remaining in the sanitary waste system after the piping cleaning has been completed. Monitor pH levels during the neutralization process to ensure effluent solution pH level is between 6.0 and 8.0.

3.3.6 Accomplish the borescope inspection of each internal surface of the CHT piping with Ship's Force upon completion of the chemical cleaning.

(V) "REPORT"

3.3.6.1 Submit One legible condition report listing results of the requirements of 3.3.6 to SUPERVISOR.

(I) "ULTRASONIC INSPECTION"

3.4 Accomplish Ultrasonic Inspection of CHT piping listed in 1.3.2 to measure wall thickness, using Paragraph 505-1.4.4 of 2.6 for guidance.

3.4.1 Wall thickness criteria must be in accordance with Paragraph 505-1.5 of 2.6.

3.4.2 Remove and install each existing pipe hanger in way of ultrasonic inspection.

3.4.3 Select and clean spot to bare metal.

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3.4.4 Measure 4 spots at 90 degree apart around circumference of piping at the following locations.

- a. Each One foot interval of straight pipe section.
- b. Each area adjacent to weld.
- c. Each of elbow and bend (Special attention must be given to the outside radius), except each silver brazed fitting.
- d. Each low point.
- e. Each location where thinning is suspected.
- f. Each area under pipe hanger.

(V) "REPORT"

3.4.5 Submit One legible condition report listing results of the requirements of 3.4 to SUPERVISOR.

3.5 Remove existing and install new piping to replace the piping listed in 1.3.3.

3.5.1 Accomplish the requirements of 009-12 of 2.1, including Table One, Column A, Lines One through 10.

3.5.1.1 Fabricate each backing ring in accordance with Line 6 for butt welding.

(V) "REPORT"

3.5.2 Submit one legible condition report of location and length of replaced piping to SUPERVISOR.

(V) (G) "OPERATING PRESSURE TEST"

3.6 Witness an operating pressure test to be conducted by Ship's Force to verify tightness of each welded joint and each mechanical joint affected by this work item and unobstructive flow, until the specified criteria is verified. Allowable leakage and obstructive flow: None. - S130

3.7 Accomplish the requirements of 009-32 of 2.1 for cleaning and preservation of each new and disturbed surface to match surrounding areas.

4. NOTES:

4.1 Initial Condition - CHT piping system is required to chemically clean and NDT inspection.

4.2 SRF tool room will issue torque wrenches when required.

5. GOVERNMENT FURNISHED MATERIAL:

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5.1 SRF will provide following materials.

MATERIAL LIST									
ITEM NO.	PC NO.	REF NO.	M / C	DESCRIPTION	STOCK NUMBER	HAZ MAT	DOC NO.	QTY REQD/ UNIT	PARA NO.
001			M	RUBBER SHEET,SOLID, 1/8 THK, 3"X3" MIL-PRF-1149, TY-2, CL-2	9B 9320 003853919	N		2/SH	3.5
002			M	O-RING FOR 3/4" IPS , UNION, M83248/1-125	9B 5331 003226397	N		10/EA	3.5
003			M	O-RING FOR 1" IPS , UNION, M83248 / 1-129	9B 5331 001675117	N		4/EA	3.5
004			M	O-RING FOR 2" IPS, UNION, M83248/1-232	9B 5331 001651950	N		4/EA	3.5
005			M	RETAINING, 3/4", IPS, RETAINER RING, FOR 3/4" IPS	9B 5325 010320947	N		10/EA	3.5
006			M	RETAINING 1", RETAINER RING, FOR 1" IPS	9B 4730 010320945	N		4/EA	3.5
007			M	RETAINING FOR 2", RETAINING RING FOR 2" IPS UNION, PG/2EA	9B 5325 010320949	N		2/PG	3.5
008			M	PIPE, 3/4"IPS, 1.050"OD X 0.065"WT, 200#, MIL-T-16420, CNA 70/30	9B 4710 000871469	N		15/FT	3.5
009			M	PIPE, 1"IPS, 1.315"OD X 0.065"WT, 200#, MIL-T-16420, CUNI 70/30	9B 4710 002774543	N		10/FT	3.5
010			M	PIPE, 1-1/4"IPS, 1.660"OD X 0.072"WT, 200#, MIL-T-16420, CUNI 70/30	9B 4710 008777417	N		10/FT	3.5
011			M	PIPE, 1-1/2"IPS, 1.900"OD X 0.072"WT, 200#, MIL-T-16420, CNA 70/30	9B 4710 013893138	N		15/FT	3.5
012			M	PIPE, 2"IPS, 2.375"OD X 0.083"WT, 200#, MIL-T-16420, CUNI 70/30	9B 4710 002897994	N		10/FT	3.5
013			M	TEE,PIPE, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014201152	N		4/EA	3.5
014			M	TEE,PIPE, LTSS, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014636170	N		4/EA	3.5
015			M	TEE, 1-1/2"IPS, SW, 400#, 803-6397430, CNA 70/30	9C 4730 SA4462248	N		2/EA	3.5
016			M	TEE,PIPE, 2"IPS, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014202544	N		2/EA	3.5
017			M	TEE,PIPE, 2"IPS, LTSS, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014636193	N		2/EA	3.5
018			M	COUPLING, COUPLING, 1/4"IPS, SW, 400#, 803-6397430, CNA 70/30	9C 4730 SA4462249	N		4/EA	3.5
019			M	COUPLING,PIPE, 1/2"IPS, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014214717	N		4/EA	3.5
020			M	COUPLING,PIPE, 3/4"IPS, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014214716	N		4/EA	3.5
021			M	COUPLING,PIPE, 1"IPS, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014214718	N		4/EA	3.5
022			M	COUPLING,PIPE, 1-1/2"IPS, SW, 400#, 803-6397430, CUNI 70/30	9B 4730 014214722	N		4/EA	3.5
023			M	COUPLING,PIPE, 2"IPS, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014221986	N		4/EA	3.5
024			M	ELBOW,PIPE, 1/4"IPS, 90, SR, SW, 6000, 545-1701701, CAN 70/30	9B 4730 011982786	N		4/EA	3.5
025			M	ELBOW,, 1/4"IPS, 90 DEG, LR, SW, 400#, 803-6397430, CNA 70/30	9C 4730 SA4462221	N		4/EA	3.5
026			M	ELBOW,PIPE, 1/2"IPS, 90 DEG, SR, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014368439	N		4/EA	3.5
027			M	ELBOW,PIPE, 1/2"IPS, 90 DEG, LR, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014397622	N		2/PG	3.5

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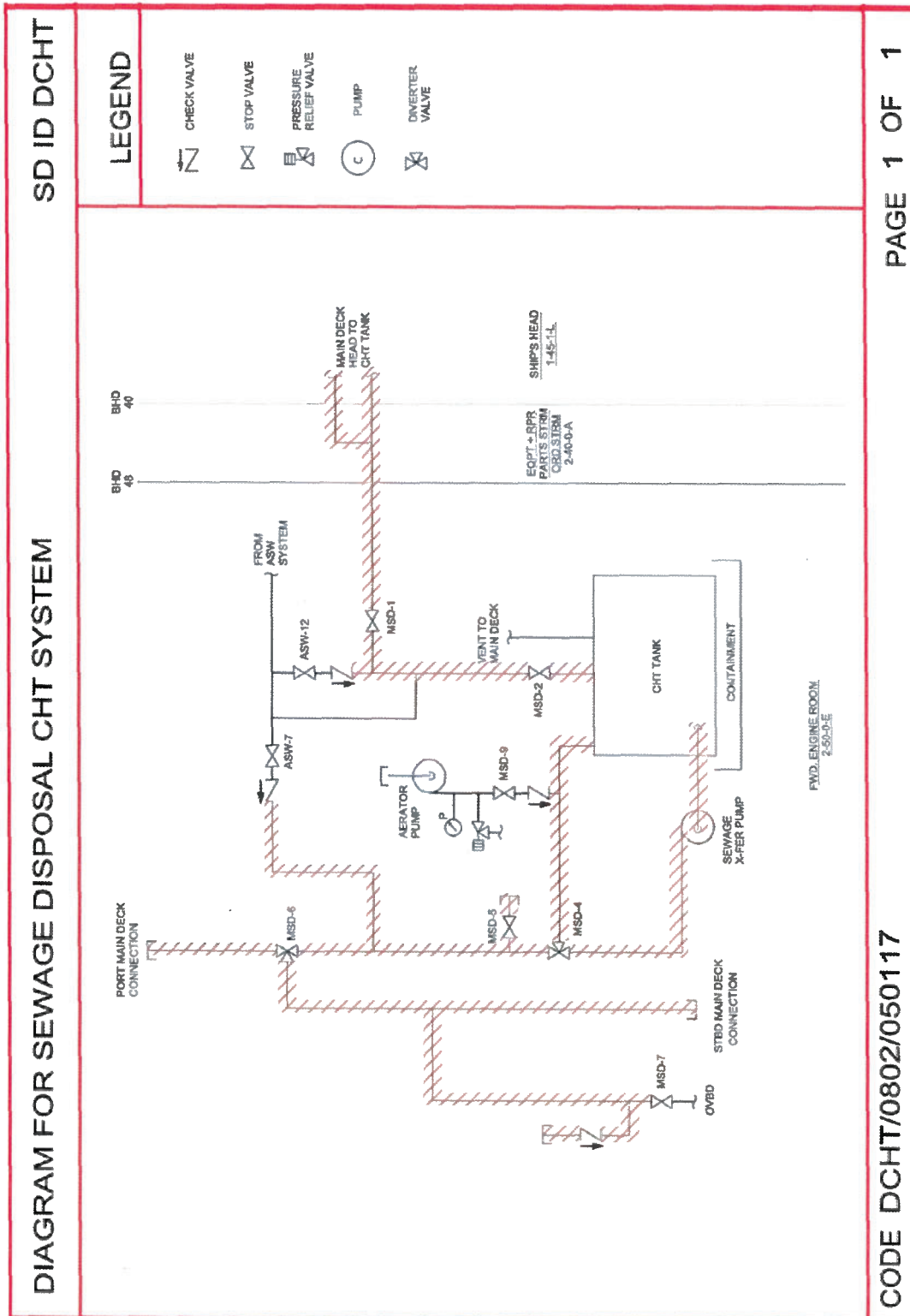
MATERIAL LIST									
ITEM NO.	PC NO.	REF NO.	M / C	DESCRIPTION	STOCK NUMBER	HAZ MAT	DOC NO.	QTY REQD/ UNIT	PARA NO.
028			M	ELBOW,PIPE, 3/4"IPS, 90 DEG, SR, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014368440	N		4/EA	3.5
029			M	ELBOW,, 3/4"IPS, 90 DEG, LR, SW, 400#, 803-6397430, CNA 70/30	9C 4730 SA4462224	N		4/EA	3.5
030			M	ELBOW,PIPE, 1"IPS, 90 DEG, SR, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014368441	N		4/EA	3.5
031			M	ELBOW,PIPE, 1"IPS, 90 DEG, LR, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014385886	N		4/EA	3.5
032			M	ELBOW,PIPE, 1-1/2"IPS, 90 DEG, SR, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014220917	N		4/EA	3.5
033			M	ELBOW,PIPE, 1-1/2"IPS, 90 DEG, LR, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014166904	N		4/EA	3.5
034			M	ELBOW,PIPE, 2"IPS, 90 DEG, SR, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014363867	N		4/EA	3.5
035			M	ELBOW,PIPE, 2"IPS, 90 DEG, LR, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014385679	N		4/EA	3.5
036			M	TEE,PIPE, 1/2"IPS, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014208659	N		4/EA	3.5
037			M	LATERAL/Y-BRANCH, 1-1/2"IPS, SW, 400#, 803-6397430, CNA 70/30	9C 4730 SA4462248	N		1/EA	3.5
038			M	LATERAL/Y-BRANCH, 2"IPS, SW, 400#, 803-6397430, CNA 70/30	9C 4730 SA4706625	N		1/EA	3.5
039			M	UNION,PIPE, UNION, 1"IPS, SW, 400#, MIL-F-1183, CNA 70/30	9B 4730 014213459	N		2/EA	3.5
040			C	UNION,PIPE, UNION AASY, 2"IPS, SW, 400#, MIL-F-1183, BRONZE NUT, CNA 70/30 BODY	9B 4730 014213449	N		1/EA	3.5
041			M	BUSHING,PIPE, BUSHING, 1-1/2" X 1"IPS, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014214671	N		2/EA	3.5
042			M	BUSHING,PIPE, BUSHING, 2" X 1-1/2"IPS, SW, 400#, 803-6397430, CNA 70/30	9B 4730 014213458	N		2/EA	3.5
043			M	ELBOW,PIPE, ELBOW, 1"IPS, 45 DEG, SW, 400#, 803-6397430, CNA 90/10	9B 4730 014635941	N		1/EA	3.5
044			C	FLANGE,PIPE, FLANGE, 2"IPS, FLAT FACE, SW, 150#, ANSI B16.5, CNA 70/30	9B 4730 015353193	N		1/EA	3.5
045			C	FLANGE,PIPE, FLANGE, 1-1/2"IPS, SW, 150#, NAVY, 810-4715319 FIG.2-A, ITEM 118, CNA 70/30	9B 4730 014635633	N		6/EA	3.5
046			M	BUSHING,PIPE, BUSHING, 2" X 3/4"IPS, SW, 400#, 803-6397430, CNA 90/10	9B 4730 016051406	N		1/EA	3.5
100			M	SCREW,CAP,HEXAGON H, 3/8"-24 UNF X 1-1/4" L, FULLY THREADED, CRES	9B 5305 015519855	N	5203B098	2/PG	3.2
101			M	NUT,PLAIN,HEXAGON, 3/8" - 24UNC	9B 5310 009031524 SX	N	5203B099	1/HD	3.2
102			M	WASHER,L, LOCK WASHER, 3/8", CRES	9B 5310 009847042 SX	N	5203B115	45/EA	3.2
103			M	RUBBER S, 24"W X 1/8"T, SYNTHETIC	9B 9320 002440192	N	5203B116	6/FT	3.2
104			M	WASHER,FLAT, 0.312"ID X 0.734"OD X 0.065"T, CRES	9B 5310 014476391	N	5203B101	100/EA	3.2
105			M	ANTISEIZE COMPOUND, 8 OZ/TU	9B 8030 002921102	Y	5226B805	1/TU	3.2

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ENCLOSURE (A) 1/1

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General Requirements
For Tank, Void and/or Cofferdam Cleaning

Ver.05012021

1. REQUIREMENTS:

1.1 Open, ventilate, conduct gas free, empty, clean, render dry, and maintain each tank, void and/or cofferdam throughout the work item performance period.

1.1.1 Ensure that each harmful vapor, fume, and mist is ventilated to the exterior of the vessel.

(V) "REPORT"

1.1.1.1 Submit one legible copy of a report listing the location, origin, and quantity of each manhole cover removed in 1.1 with respect to its tank, ship's frame and distance off centerline to SUPERVISOR after the inspection.

1.1.2 Remove and dispose of waste liquid that is not being stored for reuse to a Government provided waste liquid conveyance.

1.1.2.1 Contact the SRF-JRMC Detachment Sasebo ESH Division, Environmental Branch, S106.3 prior to transfer of waste liquid to the conveyance.

1.1.2.2 Provide watches and following spill kits (absorbent pads, booms, etc. for oil or chemical spill containment) at each affected area where waste liquid is being transferred to ensure that each liquid is not released to any shipboard compartment or the sea. In case of a spill, take immediate action to contain the spill, and contact S106.3.

1.1.2.3 Identify each transfer hose to the conveyance by affixing the company's name and tank/space location pumping from on each hose.

1.1.2.4 Establish and maintain communication between the pumping station and the conveyance when the waste liquid is being transferred.

(V) "INSPECT PIPING AND HOSE ALIGNMENT"

1.1.2.5 Inspect alignment of each hose, valve and piping in the flow path prior to the start of pumping, purging, or flushing.

1.1.2.6 Accomplish a periodic walkthrough of each piping, hose, and valve alignment for leaks at least once an hour.

1.1.2.7 Pumping after sunset is not authorized unless approved by SUPERVISOR.

ENCLOSURE (T) 1/2

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1.1.3 Coordinate the off-loading or transferring of each fluid through the ship's Damage Control Assistant (DCA), via SHIPSUPT, to maintain ship's stability and to prevent flooding.

(V) (G) "VISUAL INSPECTION"

1.2 Inspect each tank, void and/or cofferdam for cleanliness, prior to final closing. Document the personnel who were present during the inspection and confirm that they have exited the space prior to closure of each tank, void, and cofferdam. Designate one person to account for personnel who may have entered the space. -S130

1.2.1 Install each manhole cover for each tank, using each new gasket and fastener.

2. NOTES:

2.1 Booklet of General Plans and Tank Sounding Tables are available for review at the office of the SUPERVISOR.

2.2 The SUPERVISOR will provide a waste liquid conveyance to accumulate waste liquids in each tank for work being accomplished for SRF-JRMC availabilities.

2.3 See additional CHT/sewage tanks disinfection requirements of Local Standard Item 099-901YO for SRF-JRMC Sasebo availabilities.

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